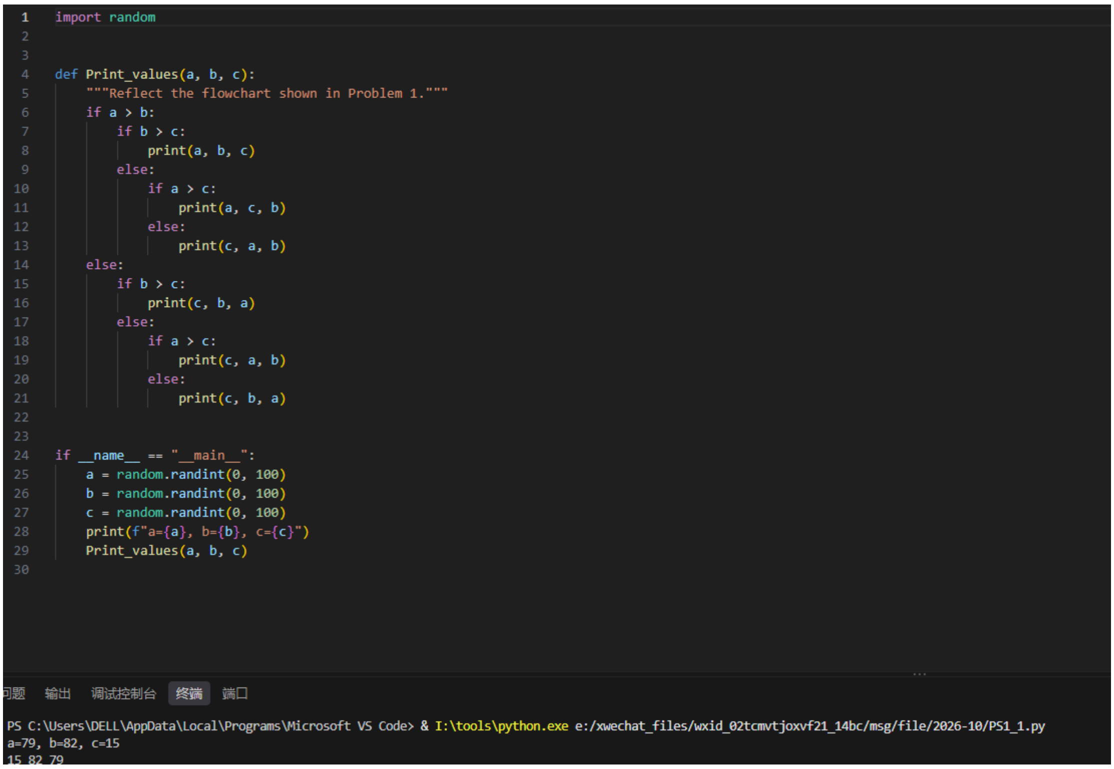
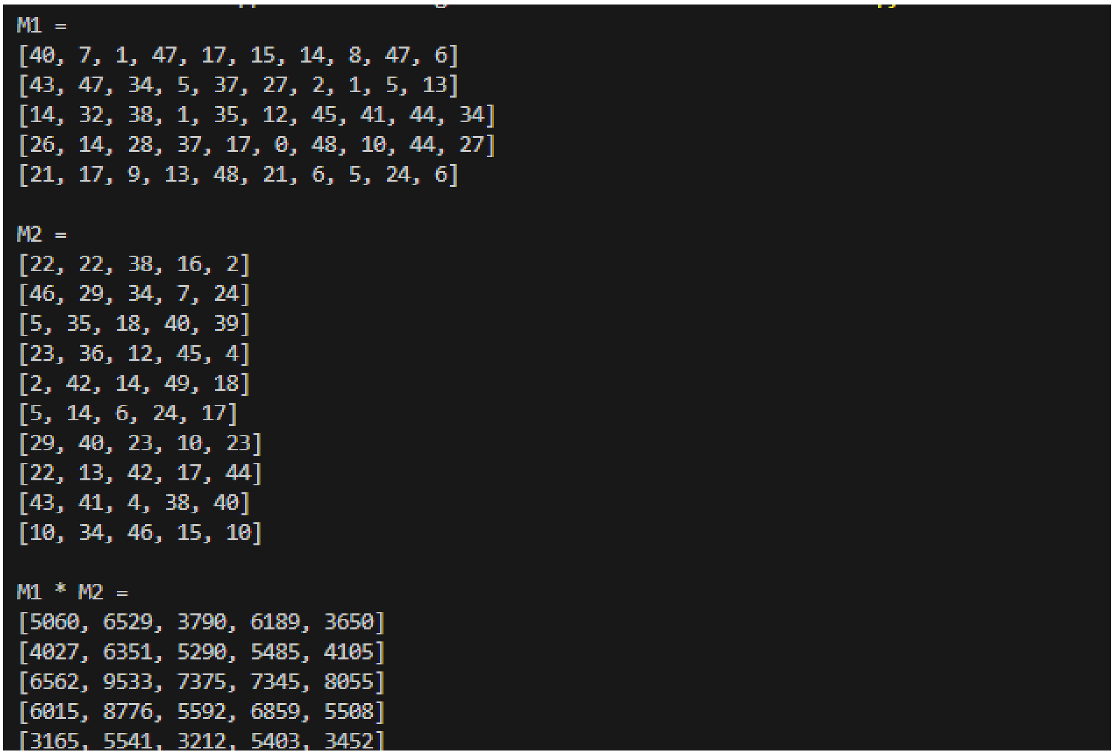
第一题



```python
import random


def Print_values(a, b, c):
    """Reflect the flowchart shown in Problem 1."""
    if a > b:
        if b > c:
            print(a, b, c)
        else:
            if a > c:
                print(a, c, b)
            else:
                print(c, a, b)
    else:
        if b > c:
            print(c, b, a)
        else:
            if a > c:
                print(c, a, b)
            else:
                print(c, b, a)


if __name__ == "__main__":
    a = random.randint(0, 100)
    b = random.randint(0, 100)
    c = random.randint(0, 100)
    print(f"a={a}, b={b}, c={c}")
    Print_values(a, b, c)

```

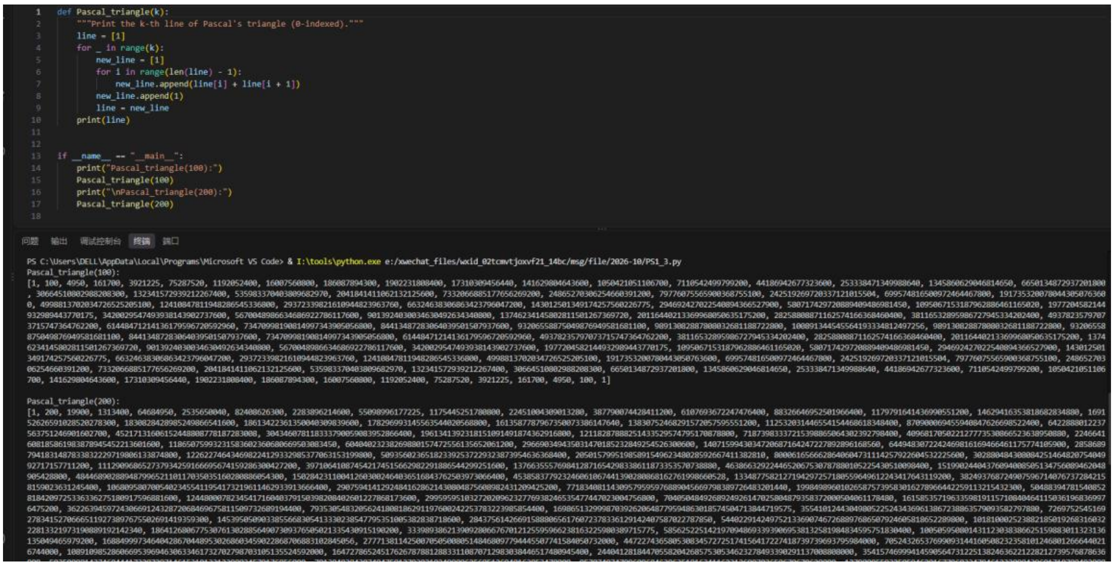
```
PS C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code> & I:\tools\python.exe e:/xwechat_files/wxid_02tcmvtjoxvf21_14bc/msg/file/2026-10/PS1_1.py
a=79, b=82, c=15
15 82 79
```

第二题



```
M1 =
[40, 7, 1, 47, 17, 15, 14, 8, 47, 6]
[43, 47, 34, 5, 37, 27, 2, 1, 5, 13]
[14, 32, 38, 1, 35, 12, 45, 41, 44, 34]
[26, 14, 28, 37, 17, 0, 48, 10, 44, 27]
[21, 17, 9, 13, 48, 21, 6, 5, 24, 6]

M2 =
[22, 22, 38, 16, 2]
[46, 29, 34, 7, 24]
[5, 35, 18, 40, 39]
[23, 36, 12, 45, 4]
[2, 42, 14, 49, 18]
[5, 14, 6, 24, 17]
[29, 40, 23, 10, 23]
[22, 13, 42, 17, 44]
[43, 41, 4, 38, 40]
[10, 34, 46, 15, 10]

M1 * M2 =
[5060, 6529, 3790, 6189, 3650]
[4027, 6351, 5290, 5485, 4105]
[6562, 9533, 7375, 7345, 8055]
[6015, 8776, 5592, 6859, 5508]
[3165, 5541, 3212, 5403, 3452]
```

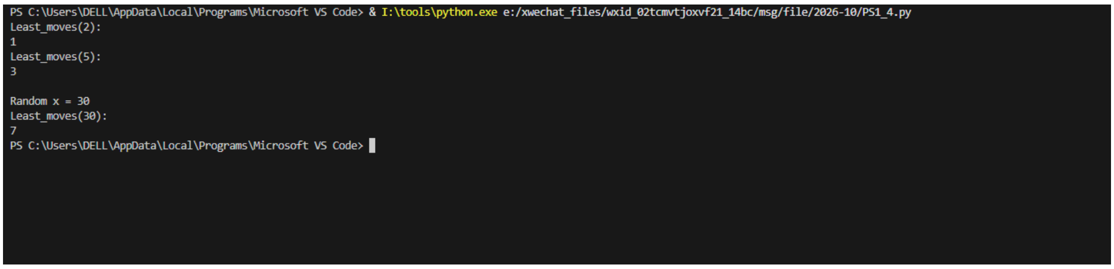
第三题



```python
def Pascal_triangle(k):
    """Print the k-th line of Pascal's triangle (0-indexed)."""
    line = [1]
    for _ in range(k):
        new_line = [1]
        for i in range(len(line) - 1):
            new_line.append(line[i] + line[i + 1])
        new_line.append(1)
        line = new_line
    print(line)


if __name__ == "__main__":
    print("Pascal_triangle(100):")
    Pascal_triangle(100)
    print("\nPascal_triangle(200):")
    Pascal_triangle(200)
```

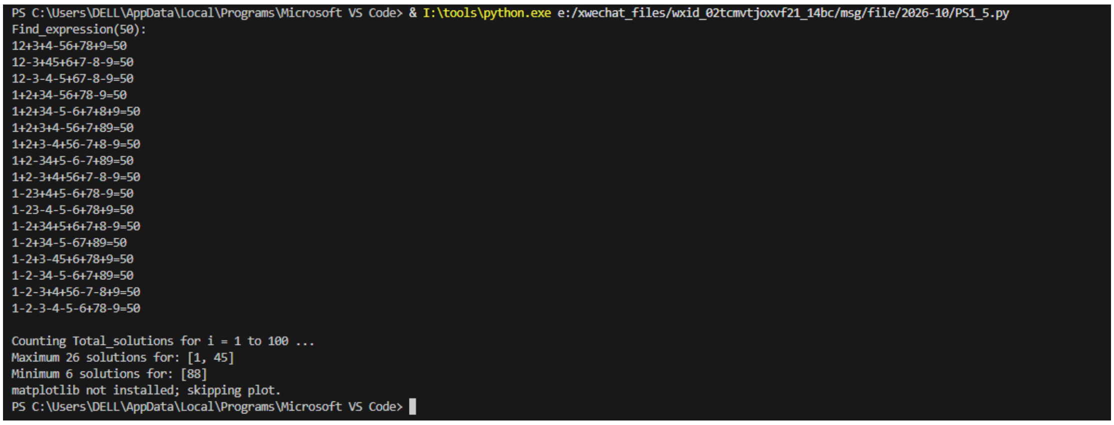
```
PS C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code> & I:\tools\python.exe e:/xwechat_files/wxid_02tcmvtjoxvf21_14bc/msg/file/2026-10/PS1_3.py
Pascal_triangle(100):
[1, 100, 4950, 161700, 3921225, 75287520, 1192052400, 16007560800, 186087894300, 1902231808400, 17310309456440, 141629804643600, 1050421051106700, 7110542499799200, 44186942677323600, 253338471349988640, 1345860629046814650, 6650134872937201800
, 30664510802988208300, 132341572939212267400, 535983370403809682970, 2041841411062132125600, 7332066885177656269200, 24865270306254660391200, 79776075565900368755100, 242519269720337121015504, 699574816500972464467800, 191735320078044305076360
0, 4998813702034726525205100, 12410847811948286545336800, 29372339821610944823963760, 66324638306863423796047200, 143012501349174257560226775, 294692427022540894366527900, 580717429720889409486981450, 1095067153187962886461165020, 1977204582144
932989443770175, 3420029547493938143902737600, 5670048986634686922786117600, 9013924030034630492634340800, 13746234145802811501267369720, 20116440213369968050635175200, 28258808871162574166368460400, 38116532895986727945334202400, 4937823579707
3715747364762200, 61448471214136179596720592960, 73470998190814997343905056800, 84413487283064039501507937600, 93206558875049876949581681100, 98913082887808032681188722800, 100891344545564193334812497256, 98913082887808032681188722800, 93206558
875049876949581681100, 84413487283064039501507937600, 73470998190814997343905056800, 61448471214136179596720592960, 49378235797073715747364762200, 38116532895986727945334202400, 28258808871162574166368460400, 20116440213369968050635175200, 1374
6234145802811501267369720, 9013924030034630492634340800, 5670048986634686922786117600, 3420029547493938143902737600, 1977204582144932989443770175, 1095067153187962886461165020, 580717429720889409486981450, 294692427022540894366527900, 143012501
349174257560226775, 66324638306863423796047200, 29372339821610944823963760, 12410847811948286545336800, 4998813702034726525205100, 1917353200780443050763600, 699574816500972464467800, 242519269720337121015504, 79776075565900368755100, 248652703
06254660391200, 7332066885177656269200, 2041841411062132125600, 535983370403809682970, 132341572939212267400, 30664510802988208300, 6650134872937201800, 1345860629046814650, 253338471349988640, 44186942677323600, 7110542499799200, 1050421051106
700, 141629804643600, 17310309456440, 1902231808400, 186087894300, 16007560800, 1192052400, 75287520, 3921225, 161700, 4950, 100, 1]

Pascal_triangle(200):
[1, 200, 19900, 1313400, 64684950, 2535650040, 82408626300, 2283896214600, 55098996177225, 1175445251780800, 22451004309013280, 387790074428411200, 6107693672247476400, 88326646952501966400, 1179791641436990551200, 14629416353818682834880, 1691
52626591028520278300, 1830828428985249866541600, 18613422361350040309839600, 178296993145563544020568800, 1613587787967350073386147640, 13830752468291572057595551200, 112532031446554154468618348400, 870900069455940847626698522400, 6422888012237
563751246901602700, 45217131606152448808778187283008, 304346078118333790059083952866400, 1961341392318151091491874362916800, 121118287888251433529574795170878800, 71873983337215398865064302392798400, 4096817050221277735308866523638950880, 2246641
608185861983878945452213601600, 11865075993231583602360680669503083450, 60404023238269880157472556135652061200, 296690349435031470185232849254526300600, 1407159943034720687164247227892896168560, 6449483072242469816169466461175774105900, 2858689
7941831487833832229719806133874800, 122622746434698224129332985377063153199800, 509356023651823392537229323873954636368400, 2050157995198589154962348028592667411382810, 8000616566628640604731114257922604532225600, 302880484300084251464820754049
92717157711200, 111290968652737934259166695674159286300427200, 397106410874542174515662982291886544299251600, 1376635557698412871654298338611873353570738880, 4638663292244652067530787880105225430510098400, 15199024404376094008505134756089462048
905428800, 48446890288948799652110117035035160280886054300, 150284231100412603002464036516843762503973066400, 453858377923246061067441390280868162761998660528, 1334877582127194297257180559649612243417643119200, 382493768724907596714076737284215
8159023631245400, 10680958070054023455411954173219611462933913666400, 290759414129248416286214308004875608982431209425200, 77183408114309579595976889045669798389726483201440, 199849896010265875739583016278966442259113215432300, 50488394781540852
8184209725336336275180917596881600, 1244800078234541716040379150398208402601227868173600, 2995959510327202096232776938246535477447023004756800, 7040504849268924926147025800487935837200050406117848 0, 1615853571963359819115710840464115036196836997
6475200, 36226394597243066912432872068469675811509732689194400, 79353054832056241808186291197600242253783223985854400, 169865132999870392620648779594863018574504713844719575, 3554101244304980052252434369613867238863579093582797800, 7269752545169
27834152706665119273897675502691419359300, 1453950509033855668305413330238547795351005382838718600, 2843756142669158880656176072378336129142407587022787850, 5440229142497521336907467268897686507924605818652289800, 101810002523882185019268316032
22813321973190889192142340, 18641268067753076130288564907309376505021335430915190200, 33398938621390928066767012125595966238163225980389715775, 58562522514219709486933939069538132581984834595751830400, 100505950001431123038386625159883011323136
135049465979200, 168849997346404286704489530268603459022868706883102845056, 277713811425007050500805148468097794445507741584050732000, 447227436580530834572725174156417227418739739693795984000, 70524326537699093144160508232358101246801266644021
6744000, 1089109852860669539694630633461732702798703105135524592000, 1647278652451762678788128833110870712983038446517480945400, 244041281844705582042685753053462327849339029113700888000, 3541574699941459056473122513824636221228212739576878636
```

第四题



```
PS C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code> & I:\tools\python.exe e:/xwechat_files/wxid_02tcmvtjoxvf21_14bc/msg/file/2026-10/PS1_4.py
Least_moves(2):
1
Least_moves(5):
3

Random x = 30
Least_moves(30):
7
PS C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code> 
```

第五题



```
PS C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code> & I:\tools\python.exe e:/xwechat_files/wxid_02tcmvtjoxvf21_14bc/msg/file/2026-10/PS1_5.py
Find_expression(50):
12+3+4-56+78+9=50
12-3+45+6+7-8-9=50
12-3-4-5+67-8-9=50
1+2+34-56+78-9=50
1+2+34-5-6+7+8+9=50
1+2+3+4-56+7+89=50
1+2+3-4+56-7+8-9=50
1+2-34+5-6-7+89=50
1+2-3+4+56+7-8-9=50
1-23+4+5-6+78-9=50
1-23-4-5-6+78+9=50
1-2+34+5+6+7+8-9=50
1-2+34-5-67+89=50
1-2+3-45+6+78+9=50
1-2-34-5-6+7+89=50
1-2-3+4+56-7-8+9=50
1-2-3-4-5-6+78-9=50

Counting Total_solutions for i = 1 to 100 ...
Maximum 26 solutions for: [1, 45]
Minimum 6 solutions for: [88]
matplotlib not installed; skipping plot.
PS C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code> 
```